Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 165 (2019) 582–590

Procedia Computer Science

www.elsevier.com/locate/procedia

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019, ICRTAC 2019

Computational Analysis of Breast Cancer Sequences using Next Generation Sequencing Methods in Julia

Babymol Kurian[a] *, V.L.Jyothi[b]

[a]Sathyabama Institute of Science & Technology, Chennai-600064, India
[b]Gurushree Shanthivijai Jain College for Women, Chennai-600064, India

Abstract

Breast cancer is one of the life-threatening diseases for mammals. Complete activities of living organisms are grounded on the DNA (Deoxyribonucleic Acid) /RNA (Ribonucleic Acid) sequences of their body. This was the salient motivation behind exploring the research on computational analysis of cancer sequences. In this paper, breast cancer sequences of Homo sapiens and mice are considered for analysis. A DNA sequence is framed with four key chemicals, A (Adenine), G (Guanine), C (Cytosine) and T (Thymine) which are referred as Nucleobases. The sequences taken for analysis vary in their size. Genome reduction for making equal sizes for all sequences may lead to non-optimal analysis and hence the original length is considered for the entire analysis. Genomic analysis such as individual Nucleobase average count, AT and GC-content, AT/GC composition, G-Quadruplex occurrence and occurrence status of Open Reading Frame (ORF) in each sequence are calculated. Execution time was calculated for sequence fetching process and ORF analysis. The human and mouse sequences were fetched approximately at 79 milli seconds(ms) and 177ms respectively. ORF analysis was processed between 640ms and 703ms. The entire analysis was done with JULIA analytical tool.

Keywords: Sequence analysis; Genomic evaluation; Julia; Open Reading Frames; G-Quadruplex

* Corresponding author. Tel.: +919884093493
E-mail address: babymolkurian@gmail.com

1877-0509 © 2019 The Authors. Published by Elsevier B.V.

10.1016/j.procs.2020.01.052

1. Introduction

Sequences are key factors in understanding the function, structure, origin, evolution, growth and distribution of living organisms. Sequence analysis play a vital role in bioinformatics to deal with DNA or RNA sequences. Various operations such as sequence alignment, sequence comparison, gene structure analysis, molecular structure analysis, frame read, mutations, genetic sequence diversity etc are common computational analysis operations applied to sequences. A DNA sequence is framed with four key chemicals, A, C, G and T which is read as AACGTATGAGC.

The genome sequencing data are available in GenBank – NCBI (National Center for Biotechnology Information). Computational analysis is applied on sequences to perform the genomic encoding prediction. The text-based sequencing data are available in various formats such as FASTA (Fast Adaptive Shrinkage Threshold algorithm), FASTQ (FAST Quality Score), EMBL (The European Molecular Biology Laboratory), GCG (Genetics Computer Group), IG (Immunoglobulins) etc. In this paper, the text data in 'fasta' are considered for analysis.

DNA has a double helical structure formed from two complementary strands of nucleotides held together by hydrogen bonds between G-C and A-T base pairs (dual sequences). Forward/Positive (+) and Reverse/Negative (-) strands are considered for the analysis. Out of the dual strands, either forward or reverse sequence is considered for analysis. The digitized form of forward strand is referred as the original sequence, which is mostly preferred for analysis. Plethora of research is carried out in DNA sequencing. Supplementary information such as the gene location in the sequence, nucleotide order and their functions, interaction among the genes etc can be inferred through sequence analysis.

The concept of sequence analysis can be extended to chemical industry for framing the polymer with several monomers, finding the Next Product to buy in marketing, life-career path, work-life balance in sociology and so on.

2. Literature Survey

Various protein encoding genes of chloroplast genomes of different species of Nicotiana and Solanum was considered for study in [1]. AT%, GC% and AT/GC composition was calculated for all species. It was concluded that the base compositions in the plastomes were almost unvarying in their GC%. In all species under study, these ratios and percentages were found to be consistent. The reason may be because they belong to the same taxonomic group or due to less digression in the plastomes. This was because species taken for study were at the family level only. The paper [2] focused on the genomic instability due to cancer. Computational approaches to reveal mechanisms underlying cancer mutation were studied. The influence of non-B DNA structure symbolized by G4 DNA sequences is surveyed computationally and observed to effect transcription, transcription and replication directly in genome instability and cancer. It was observed that G4 DNA-forming sequences were enhanced two-fold at translocation breakpoints and led to genomic variability in cancer. The output point to G4 DNA was one of the factors for the formation of mutations in cancer cells.

Guanine-rich DNA or RNA sequences comprised of G-quadruplexes (G4s) and nucleic acid secondary structures which were well explained in [3]. Chromatin architecture and gene regulation were affected by G4 formation and were associated with genetic diseases, genomic instability and cancer progression. In the paper, a high-resolution sequencing–based method to distinguish G4s in the human genome was offered. Oncogenes, tumour suppressors and somatic copy number alterations interrelated to cancer development were highly allied with G4 formation. In [4], detection and mapping of G4 structures in the human genome and in biologically relevant contexts were progressed recently to an extent. G4 structures control transcription and genome stability, and revealed their potential relevance for cancer therapy. G4 was formed throughout the human genome and in gene regulatory regions, which mostly approved previous computational predictions. This paved the way for the biological significance of G4 structures, signifying that they had roles in gene regulation and genomic instability, particularly in cells with specific genetic deficiencies.

The paper [5] presented a comprehensive examination of smORFs(small ORF) in flies, mice and humans. It has proposed the existence of ample functional classes of smORFs, ranging from inert DNA sequences to transcribed and translated cis-regulators of translation and peptides with an inclination to function as regulators of membrane-associated proteins. Steps in gene, peptide and protein evolution were represented by different smORF classes. The paper familiarized a distinction between different peptide-coding classes of smORFs in animal genomes, and high pointed the role of model organisms for the learning of small peptide biology in the context of development, physiology and human disease. Certain working rules to foresee which primary sequences can form Quadruplexes and a search algorithm is elucidated in [6]. There was a significant suppression of quadruplexes in the coding strand of exonic regions, which recommends that quadruplex-forming patterns were disapproved in sequences that would form RNA. G-quadruplex is a possible nucleic acid-based mechanism for regulating multiple biological processes [7]. The sequencing of many genomes has discovered that they are rich in sequence motifs that have the potential to form G-quadruplexes and that their location is non-random, correlating with functionally important genomic regions. The support vector machines were used to tackle the fold classification problem for a target sequence and the alignment of hidden Markov models [8]. Using the consensus of many fold recognition methods remains to be one of the most successful strategies for both recognition and alignment of remote homologues.

3. Digitized data and tools

3.1. Sequence Structure

The analysis performed to find the precise order of nucleotides, adenine (A), guanine (G), cytosine (C) and thymine (T) in a given DNA sequence is referred as sequencing [9].

```
>NZ_AHCB02000001.1 Pseudoalteromonas marina DSM 17587 strain mano4 contig001, whole genome shotgun sequence
GCTCCTTGTCTTCTCTGTGGTGATAATAATGTTTTATTCACTTTGAAACTTTTTACAGCTTGTATCCATAAGGGCAGTTTTTACAGCCATTTTTACAGCAGGTTCCT
CGCTTTAAAAAAAACCACTTACTAAATACCATGTTGCCATTTTGCATGGTGTAGTCGAGTTCTTCAATTAAGTTATCGTTTCCTTTAAATGCTTTTGCAAACGCAAT
TTGATCTTTAATTGGTTGGGTATATAGCTCTTCAAGGTAGTGATTAATTTTATTGAGTGTACAGGTTTGGCACAAACAGCTTGTTGCAGCTGCGTCTAGGGGTAAAA
TAGCAGGTAACTGGTTACACCAACATGCGGTTATATTATTTACATTACAGTTAAGTACTGTATTACATTGGCTGCAGTTTGTTTGAGTCATTTTAGTTCTTATCAAC
AGAAGAGCGCCAAGAATAACAAATATTATGAATATACAAAAAGCCTTAGCAGAATTTATTACGTTTTCCGCACAAAGAGACGATGCAATATCGGTTATTGTGTCAAC
```

Fig. 1. ‘FASTA’ file structure

Two different types of data sets, Homo sapiens and mice with Cancer disease are considered for computation. The required data are fetched from the GenBank online web link [10]. DNA sequence in FASTA format is given in Fig. 1. A sequence in FASTA format commences with ‘>’ symbol followed by a single-line description and trailed by lines of sequence data.

3.2. Software Tool – Julia-Version 1.1.1

The enhanced features like exception handling, binary operation support and macro applications are introduced in the new version of Julia. Older versions of Julia are not sufficient for analyzing the biological sequences represented in ‘fasta’ text form. A ‘Bio.Seq’ module is used in addition to Julia for supporting the computational sequencing tasks. The methods and functions defined in this module make use of ‘fasta’ file for sequence extraction and graph generation for fetched sequences.

4. Sequence Analysis

Five partial sequences with variable sizes were taken for Homo sapiens and mouse with cancer disease. The components of sequence analysis are the fasta file as input, Bio-Julia module for making the sequencing analysis, which comprised of sequence fetching, sequence length detection, average individual genomic occurrence calculation, AT and GC content detection, AT/GC ratio, G - Quadruplex occurrence and Open Reading Frame (ORF) status and the various computational information such as Genomic ranges, AT/GC ratio chart, G-Quadruplex values and ORF status as output. Architectural diagram for the sequencing analysis is shown in Fig. 2.

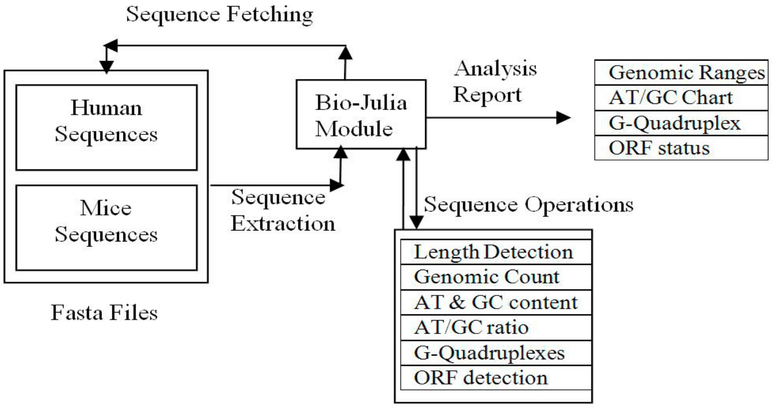

Fig. 2. Diagram of Sequence Analysis

4.1. Sequence Fetching Process

Fasta file contents are extracted using the methods in ‘Bio.Seq’ module in Julia. The processing steps with Julia commands are given in Table 1. For loop in Julia has, ‘desc’ and ‘seq’ variables. These variables are used to store the description details of sequence in the Fasta format. ‘push () function is used to insert the fetched sequence into a stack type storage variable ‘x’.

Table 1. Sequence Fetching Process

Algorithmic Steps	Julia Implementation
Include Bio.Seq module in the coding segment	Using Bio.Seq
Include ‘FastaIO’ module for fasta file process	Using FastaIO
Define a variable ‘x’ of vector type to hold the sequence	X=Vector(undef,0)
Open the Human sequence fasta file and assign a file pointing variable ‘fr’ to represent the file	FastaReader(“human.fasta”) do fr
Apply ‘for’ loop in extracting sequences from the file using the file pointer ‘fr’.	For (desc,seq) in fr push! (x, seq) end
Terminate the file pointer ‘fr’	end
Print the total number of sequences in the fasta file	println(length(x))

4.2 Sequence Length Detection

The sequences taken for the analysis were partial sequences with unequal size. The length of every sequence is required to find the average analysis. It is identified using the following Julia coding segment. The sequences in the stack variable ‘x’ are read one at a time and the current sequence is placed in the variable ‘i’. length() function is used to find the size of the current sequence in ‘i’.

Julia Code:

```
for i in x
print(length(i),” “)
end
```

4.3 Genomic Count

The total occurrence of each genomic element is calculated in the equation (1).

$$C_g = \frac{\sum_{i=1}^{S_{len}} g}{S_{len}} \tag{1}$$

Where

C_g Count of the particular genome, g.

g Values A, G, C and T

S_{len} Length of the Sequence

4.4 AT and GC content

GC content is measured by the summation of G and C divided by the total length of the sequence and AT content is measured by the summation of A and T divided by the total length of the sequence, shown in equations (2) & (3). These measures are used for finding out the sequence stability.

$$GC = \frac{C_G + C_C}{S_{len}} \tag{2}$$

$$AT = \frac{C_A + C_T}{S_{len}} \tag{3}$$

4.5 AT/GC measure

AT/GC ratio, R is calculated as the sum of the average occurrence of the genomes, A and T divided by the sum of the average occurrence of the genomes G and C in a sequence, shown in equation (4).

$$R = \frac{AT}{GC} \tag{4}$$

4.6 G-Quadruplex occurrence

This analysis is required to evaluate the association with disease acquisition [11]. Four contiguous occurrence of ‘G’ nucleotide in the sequence is identified by using the Julia code in Fig. 3. The code is used to find the total occurrence of G-Quadruplex in each sequence.

```
for i in x
   s1=String(i)
   gene1="GGGG"
   r = Regex(string(gene1))
   l = [m. match for m = eachmatch (r, s1)]
push! (repent, length(l))
end
```

Fig. 3. G-Quadruplex count detection

4.7 Open Reading Frame (ORF)

The sequential segment between the start codon and stop codon is referred as an Open Reading Frame (ORF). Existence of the ORF(s) is an indication for the protein occurrence in the sequence. If there is an existence of the protein, it may be cancer inducing or protecting. To find the ORF, the sequence should be split into a set of three nucleotides referred as codons, shown in Fig. 5. ORF existence is identified by the subsequence between the start codon “ATG” and any one of the end codons “TAA”/” TAG”/”TGA” [12].

Table 2. Codon Generation

Input Sequence in Julia	GGGGGGGTGACTCTGTATCTCGGTGAGGAGCACGTTCTTCTGCTGTATGT AACCTGTCTTTTCTATGATCTCTTTAGGGGTGACCCAGTCTATTAAAGAA AGAAAAATGCTGAATGAGGTAAGTACTTGATGTTACAAACTAACCAGAG ATATTCATTCAGTCATATAGTTAAAAATGTATTTGCTTCCTTCCCTCAATG CACCACTTTCCTAAAACAATGAAGACA
Codon generation from Julia	["GGG", "GGG", "GTG", "ACT", "CTG", "TAT", "CTC", "GGT", "GAG", "GAG", "CAC", "GTT", "CTT", "CT "GTA", "ACC", "TGT", "CTT", "TTC", "TAT", "GAT", "CTC", "TTT", "AGG", "GGT", "GAC", "CCA", "GTC", "T "AAG", "AAA", "AAT", "GCT", "GAA", "TGA", "GGT", "AAG", "TAC", "TTG", "ATG", "TTA", "CAA", "AC "AGA", "TAT", "TCA", "TTC", "AGT", "CAT", "ATA", "GTT", "AAA", "AAT", "GTA", "TTT", "GCT", "TCC", "T "TGC", "ACC", "ACT", "TTC", "CTA", "AAA", "CAA", "TGA", "AGA", "CA"]

5. Results and Discussion

Two different sets of human and mouse sequences with cancer dieses were analyzed. In each category, five sequences were considered for analysis. The lengths of two sets of sequences are given in Table 3. Average genomic occurrence in the sequence is evaluated for both sets of data and the outcome is shown in Fig. 4.

Table 3. Sequence Length

	Sequence Length				
	S1	S2	S3	S4	S5
Human	222	227	228	246	226
Mouse	511	1386	1385	1386	370

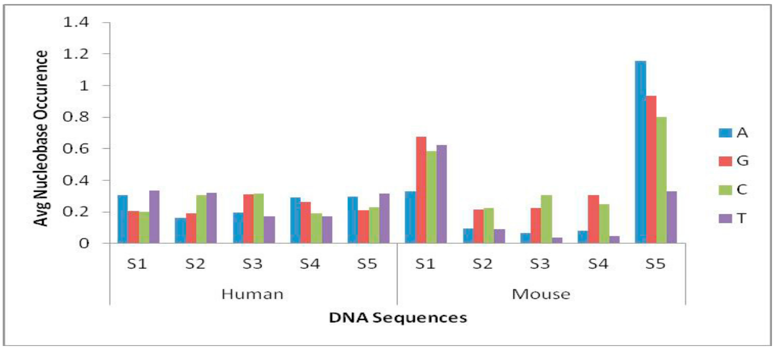

Fig. 4. Average genomic occurrences

The nucleotide range for human and mouse are depicted in Table.4. With respect to the minimum average occurrence, the value of ‘A’ and ‘T’ of human is greater than 0.1, but for mouse it is less than 0.01. The minimal range of ‘G’ and ‘C’ in human is less than 0.2 but for mouse it is greater than 0.2. The maximum range of all four nucleotides in human varies from 0.30 to 0.34, but in mouse the values are: ‘A’ with 1.15676, ‘T’ with 0.62231, G with 0.93513 and T with 0.8027, which is almost greater than .62. Fig. 5 shows the AT and GC content which are evaluated from the average occurrences of the nucleotides.

Table 4. Nucleotide range

Nucleotide	Human		Mouse	
	Min	Max	Min	Max
A	0.16299	0.30631	0.06931	1.15676
G	0.18943	0.31141	0.21717	0.93513
C	0.19106	0.31579	0.22511	0.8027
T	0.17105	0.33783	0.03682	0.62231

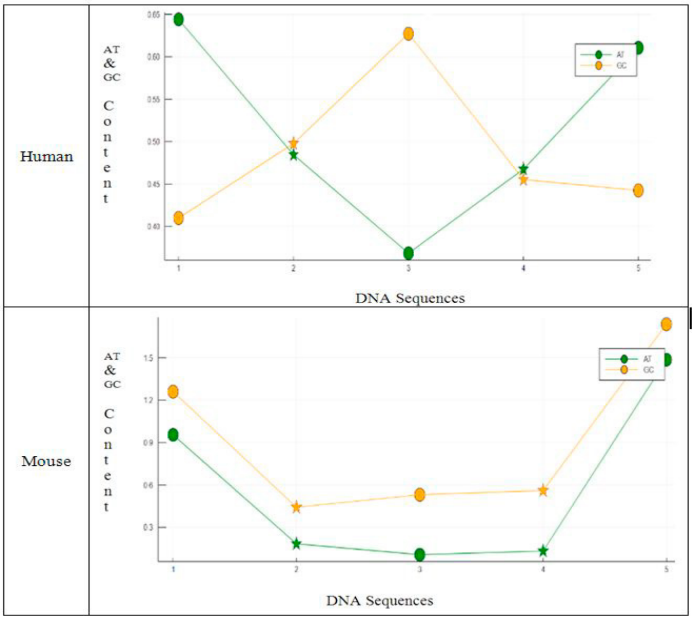

Fig. 5. AT & GC Content

AT/GC ratios are also evaluated from the average occurrences shown in Fig. 6. From the Figure, it is clear that for Human beings, ratios are ranging between 0.58 and 1.56, but for mouse, ratios are between 0.2 and 0.86.

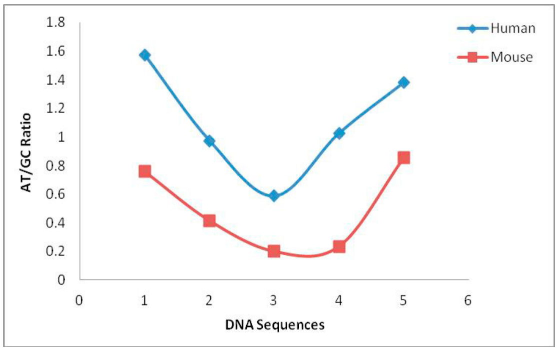

Fig. 6. AT/GC Ratio

G-Quadruplex is extracted from 5 sequences. All sequences have minimum of 1 4G occurrence and maximum of 3 4G occurrence is found in mouse. The existence of G-Quadruplex is an indication for the presence of cancer in an organism. The G-Quadruplex (4G occurrences) count along with the sequence length is given in Table 5. ORF analysis is shown in Table 6. ORF is calculated for both sequences to know the possibility of protein existence in the sequence. A stretch of sequence which commence with the start codon and end with the stop codon, indicates the presence of an ORF protein. The start codon and the stop codon positions are listed along with the total number of nucleobases between the start and stop codon as the ORF length. The table also carries additional information such as total number of ORF proteins in each sequence. Sequence 3 (S3) of human has two ORF, the first ORF starts at position 39 and ends at 41 whereas the second ORF starts at position 67 and ends at 72, with the total ORF count as 2. In the same way, S4 and S5 of human have two ORF. But, in mouse, the sequences, S2, S3 and S4 have start codons at 9 positions, without even a single end codon, which indicates, nil protein existence in the sequence. From the table, it is clear that, almost all human sequences have minimum of one ORF protein, but, in mouse, all three sequences have no ORF protein existence.

Table 5. G-Quadruplex count

	S1		S2		S3		S4		S5	
	Length	4G Count	Length	4G Count	Length	4G Count	Length	4G Count	Length	4G Count
Human	222	1	227	2	228	2	246	2	226	1
Mouse	511	2	1386	3	1385	3	1386	2	370	1

The processing time for fetching all 5 sequences from each file is evaluated using tick () and tock () functions, encapsulated in 'TickTock' module. The execution time is estimated in milliseconds. The execution speed is shown in Table 7. Processing time is also calculated for ORF process, shown in Table 8.

Table 6. ORF analysis

.	Human					Mouse				
	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
ORF Length	74	76	76	82	76	171	462	462	462	124
Start Codons	38	44	39/67	39/67	17/23	28	9 pos	9 pos	9 pos	16/93
Stop Codons	40	74	41/72	41/72	18/44	55	-	-	-	71/97
ORF Count	1	1	2	2	2	1	0	0	0	2

Table 7. Execution time for sequence Fetching Process

Sequence Set	Fetching Time
Human	79 ms
Mouse	177 ms

Table 8. Execution Time for ORF Analysis (in milliseconds)

Sequence Set	ORF Analysis – Execution Time				
	S1	S2	S3	S4	S5
Human	679	641	684	684	666
Mouse	665	703	632	663	658

6. Conclusion

Gene Sequence analysis is a computational research area with much scope in understanding the basic features, structures and functionalities of the living organism. In this paper, 5 formatted breast cancer sequence files (BRCA1-Breast Cancer type 1) for Human and mouse were considered. The computational analysis such as finding the length of the sequence, average of each nucleotide presence in all sequences, AT & GC content, AT/GC ratios, G-Quadruplex occurrences and ORF occurrences were evaluated using Bio.Seq module in Julia 1.1.1 data analytical tool. G-Quadruplex occurrence varies from 1 to 3, which indicates, disease acquisition (cancer). AT/GC ratio for human being ranges between 0.58 and 1.56, but for the mouse, the ratio varies between 0.2 and 0.86. The execution speed for human sequence fetch is observed lesser than the mouse. The analysis can be extended to apply on various species of organisms and formulate the comparison among them. The computational analysis for BRCA2(Breast Cancer type 2) sequence files for various species can be conducted. The comparative computational analysis between various types of cancer sequences can be done for future work.

References

[1] Sharma, B.C., Ghimiray, D. and Chhetri, A. (2013) "Study of the AT/GC composition in chloroplast genomes of Solanaceae shows strong conservation of base-composition in genes of plastomes." *Journal of Scientific and Innovative Research* ***2*** **(6)**: 1017-1022.
[2] Bacolla, A., Ye, Z., Ahmed, Z. and Tainer, J.A. (2019) "Cancer mutational burden is shaped by G4 DNA, replication stress and mitochondrial dysfunction" *Progress in biophysics and molecular biology*.
[3] Chambers, V.S., Marsico, G., Boutell, J.M., Di Antonio, M., Smith, G.P. and Balasubramanian, S. (2015) "High-throughput sequencing of DNA G-quadruplex structures in the human genome." *Nature biotechnology* ***33*****(8)**: 877.
[4] Hänsel-Hertsch, R., Di Antonio, M. and Balasubramanian, S. (2017) "DNA G-quadruplexes in the human genome: detection, functions and therapeutic potential." Nature *reviews Molecular cell biology* ***18*****(5)**: 279.
[5] Couso, J.P. and Patraquim, P. (2017) "Classification and function of small open reading frames." *Nature reviews Molecular cell biology* ***18*****(9)**: 575.
[6] Burge, S., Parkinson, G. N., Hazel, P., Todd, A. K., & Neidle, S. (2006). Quadruplex DNA: sequence, topology and structure. *Nucleic acids research*, ***34*****(19)**, 5402-5415.
[7] Rhodes, D., & Lipps, H. J. (2015). G-quadruplexes and their regulatory roles in biology. *Nucleic acids research*, ***43*****(18)**, 8627-8637.
[8] Dunbrack Jr, R. L. (2006). Sequence comparison and protein structure prediction. *Current opinion in structural biology*, ***16*****(3)**, 374-384.
[9] https://towardsdatascience.com/dna-sequence-data-analysis-starting-off-in-bioinformatics-3dba4cea04f
[10] https://www.ncbi.nlm.nih.gov/nuccore/?term=Homo+sapiens+breast+cancer+type+1+susceptibility+protein)
[11] Wang, Y., Yang, J., Wild, A.T., Wu, W.H., Shah, R., Danussi, C., Riggins, G.J., Kannan, K., Sulman, E.P., Chan, T.A. and Huse, J.T.(2019) "G-quadruplex DNA drives genomic instability and represents a targetable molecular abnormality in ATRX-deficient malignant glioma. "*Nature communications* ***10*****(1):** 943.
[12] https://vlab.amrita.edu/index.php?sub=3&brch=273&sim=1432&cnt=1